

KPP Validation Report

MITgcm vs Python Port Comparison

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MITgcm Reference:

File: mitgcm_kpp_outputs_1D_10_kppmix_extend_rawflux_fix.nc
UUID: 6d56ff34-f8ff-4406-81ee-d73f5f3d6162

Python Port Output:

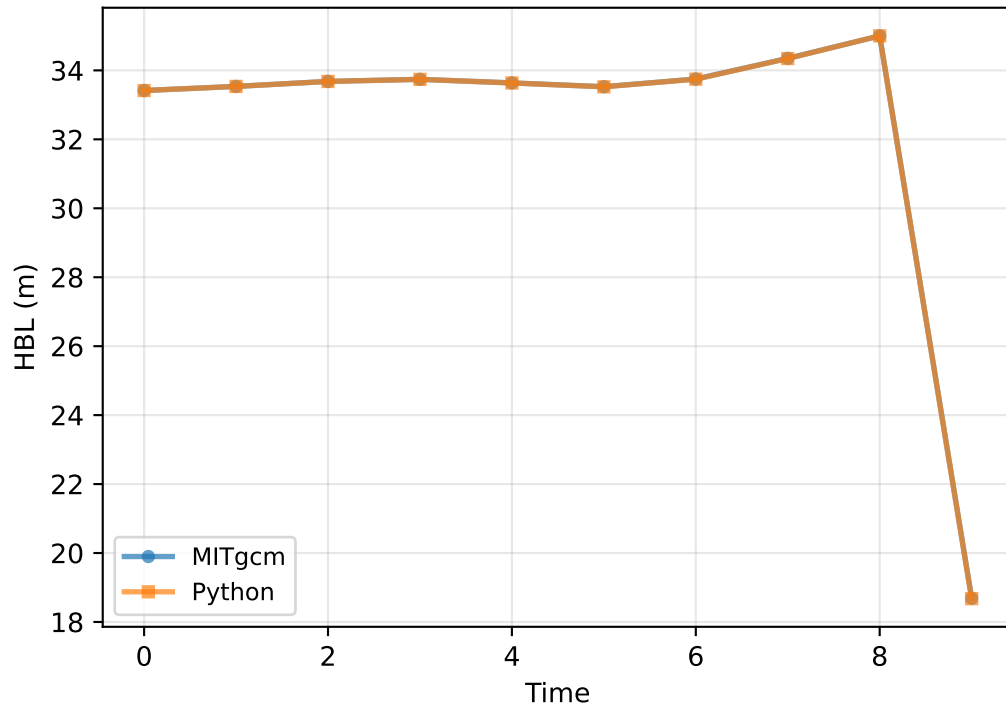
File: python_kpp_outputs_1D_10_kppmix_extend_rawflux_fix.nc
UUID: 6d56ff34-f8ff-4406-81ee-d73f5f3d6162

Provenance: ☐ UUIDs match

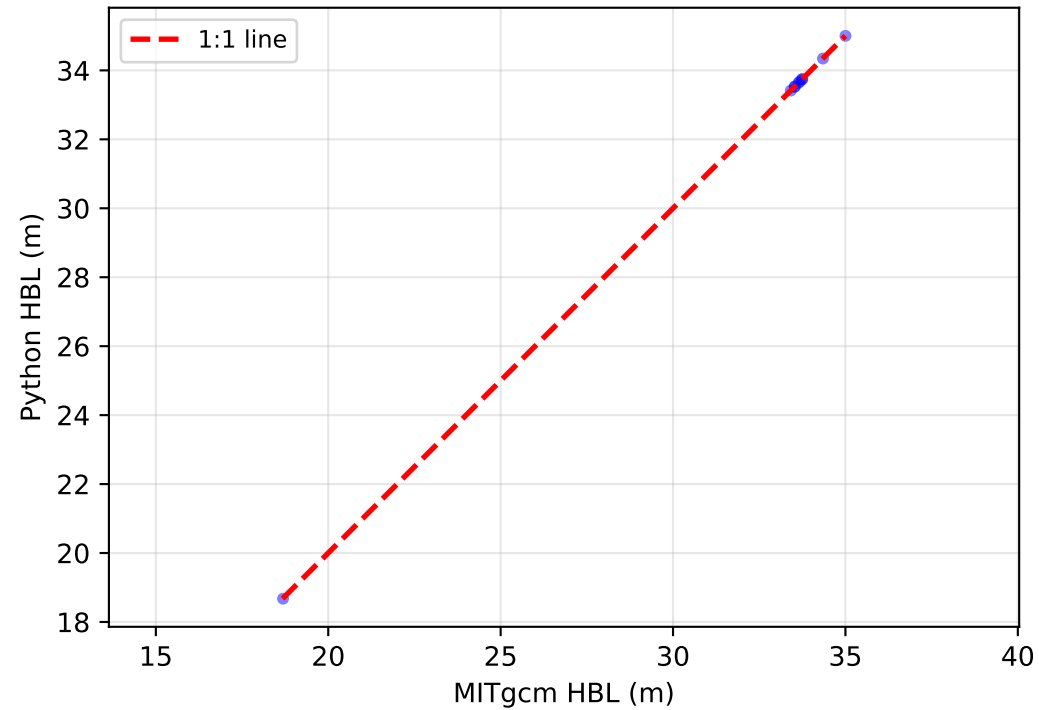
Timesteps: 10
Grid: 1 × 1 × 23

Boundary Layer Depth (HBL) Comparison

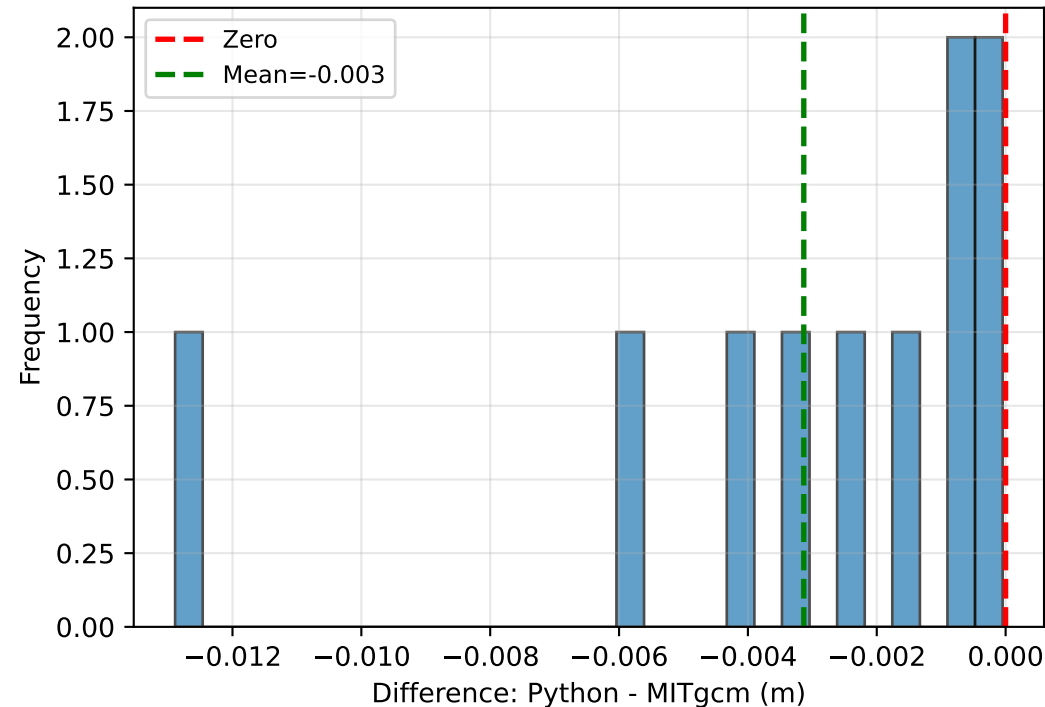
HBL Time Series



HBL Scatter Plot



HBL Difference Distribution



HBL Statistics:

Mean |diff|: 0.003133 m
Max |diff|: 0.012891 m
RMS diff: 0.004834 m

Mean rel error: 0.0123%
Median rel error: 0.0054%

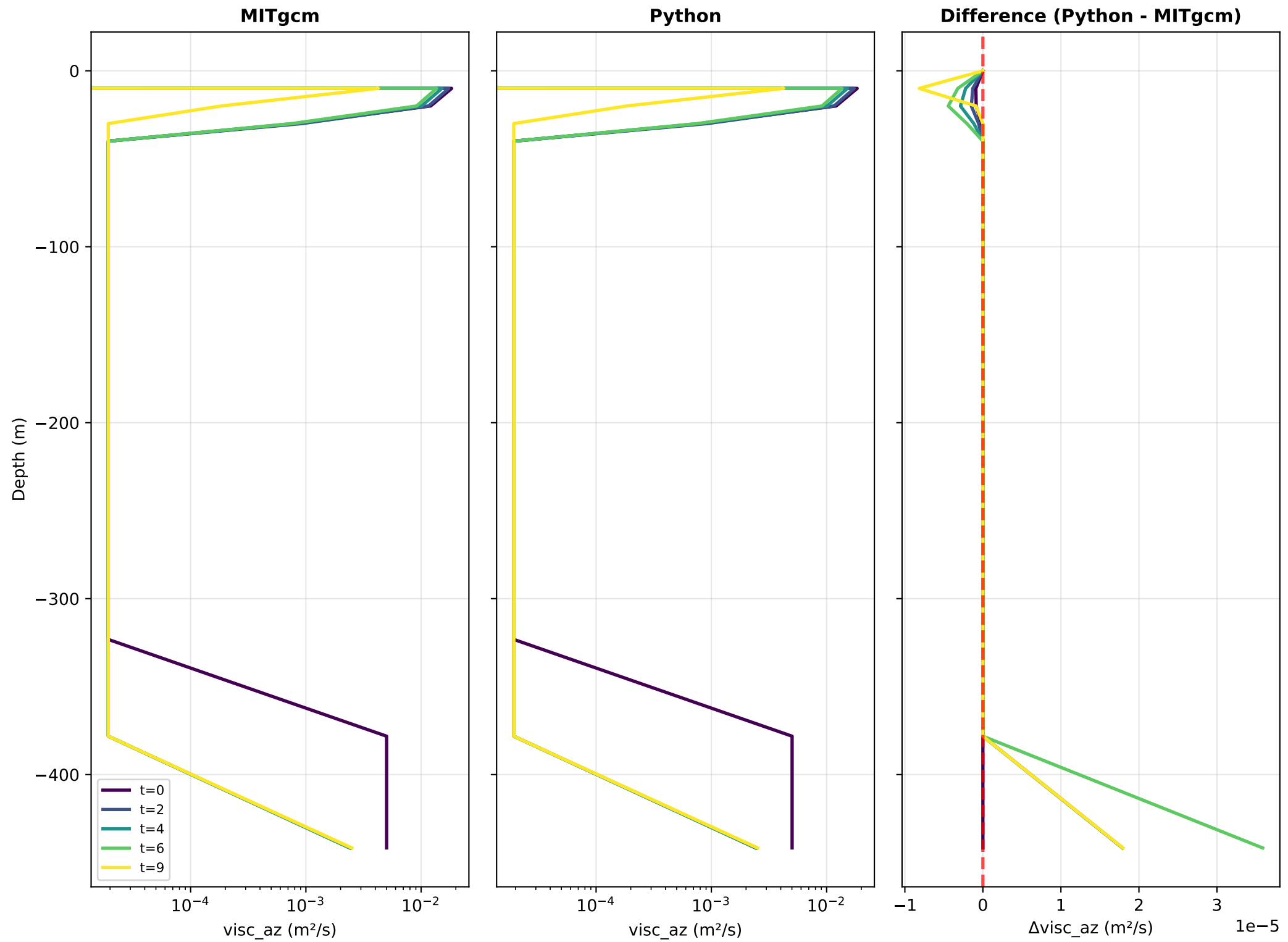
Points compared: 10

Range:
MITgcm: 18.7 – 35.0 m
Python: 18.7 – 35.0 m

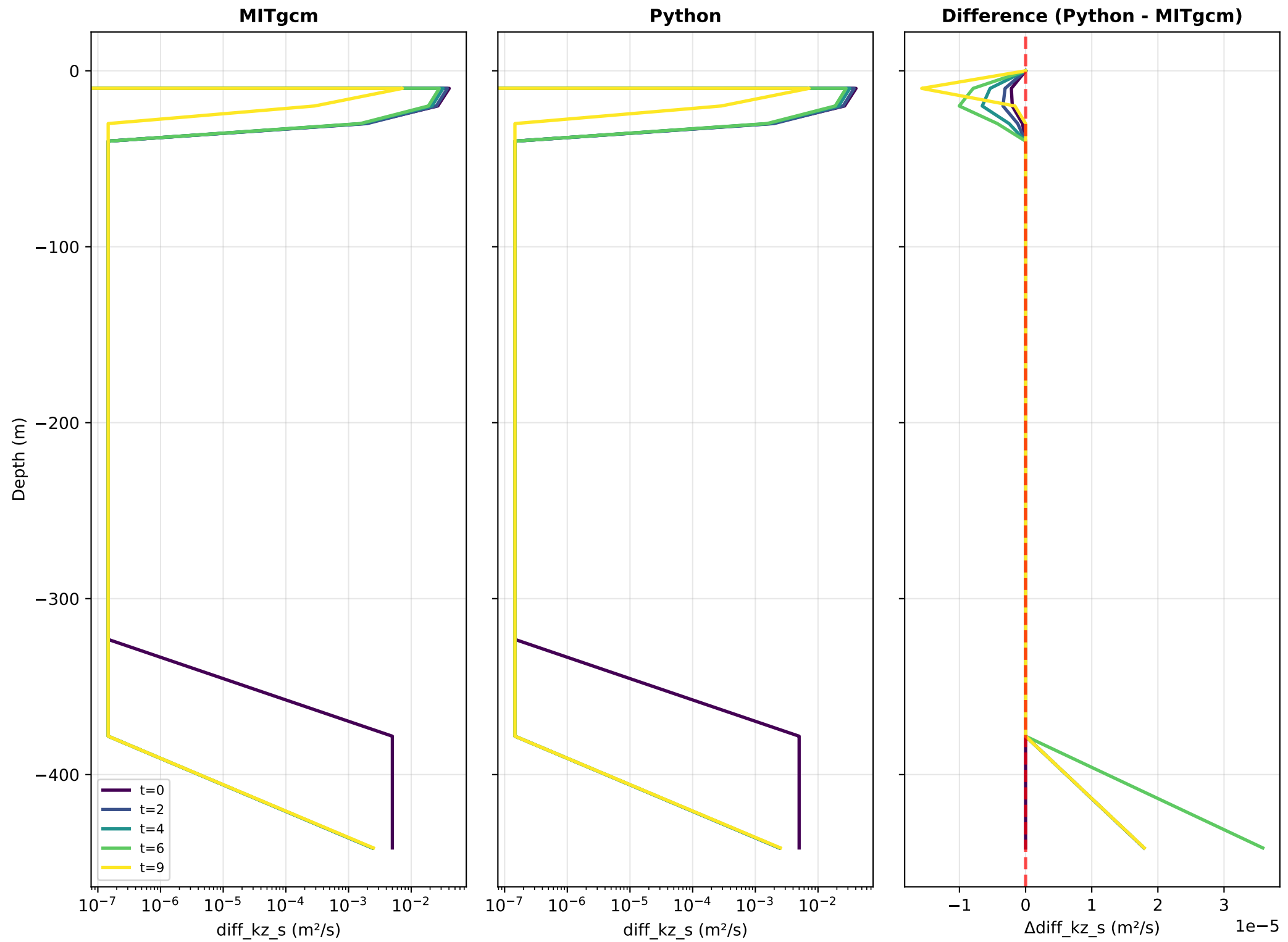
Assessment:

EXCELLENT

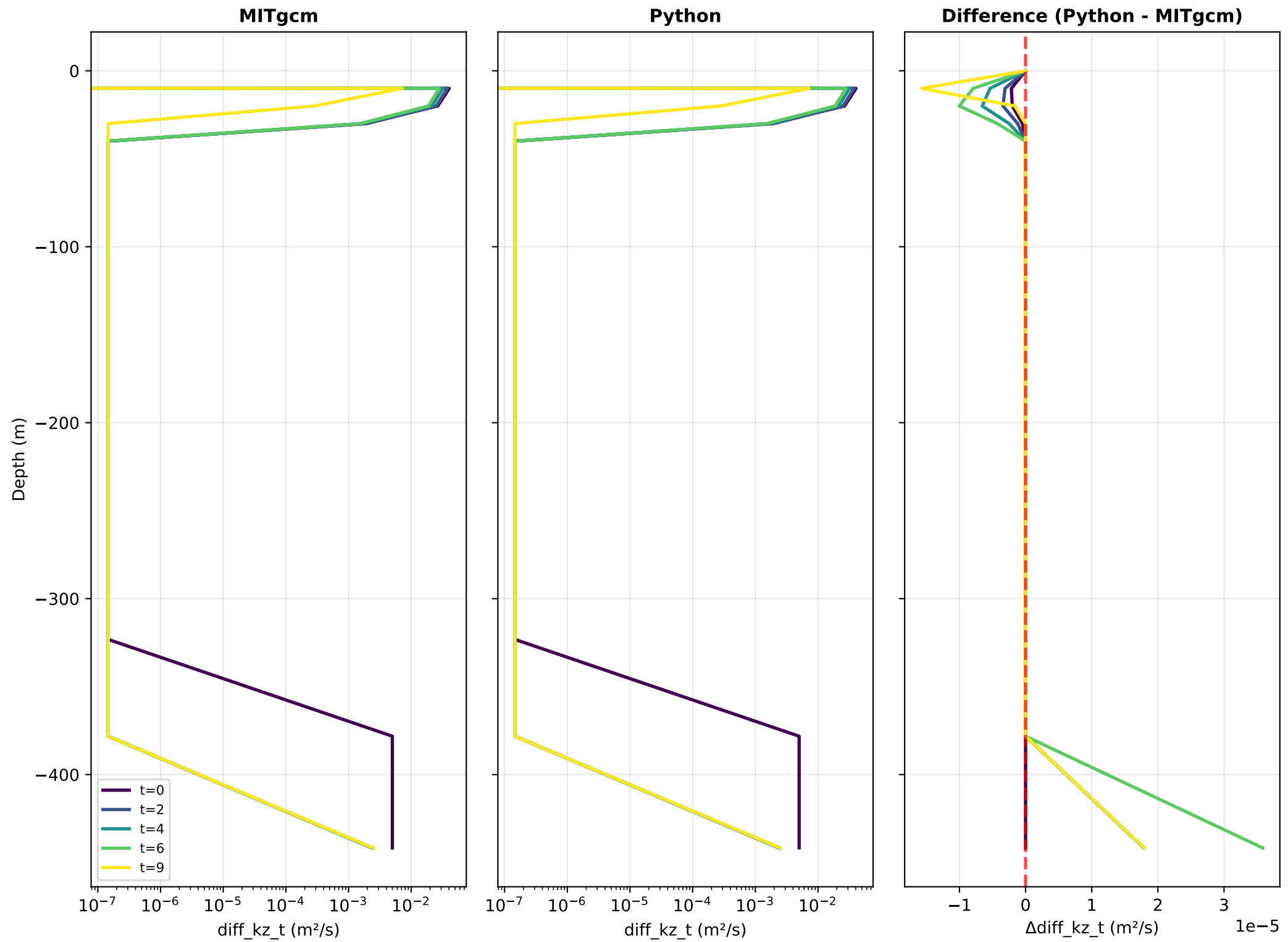
VISC_AZ Vertical Profiles



DIFF_KZ_S Vertical Profiles



DIFF_KZ_T Vertical Profiles



Mixing Coefficients Statistics

Active Mixing Regions (coefficient > 1e-6 m²/s)

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VISC_AZ: □ EXCELLENT

Median relative error:	0.000000%
Mean relative error:	0.043477%
Max relative error:	1.461992%
95th percentile error:	0.208271%
Median absolute diff:	0.000000e+00 m ² /s
Points compared:	220

DIFF_KZ_S: □ EXCELLENT

Median relative error:	0.037038%
Mean relative error:	0.245251%
Max relative error:	1.473498%
95th percentile error:	0.787811%
Median absolute diff:	4.157459e-06 m ² /s
Points compared:	40

DIFF_KZ_T: □ EXCELLENT

Median relative error:	0.037038%
Mean relative error:	0.245251%
Max relative error:	1.473498%
95th percentile error:	0.787811%
Median absolute diff:	4.157459e-06 m ² /s
Points compared:	40

GHAT: □ EXCELLENT

Median relative error:	0.012150%
Mean relative error:	0.020643%
Max relative error:	0.127815%
Points compared:	28

Validation Summary & Assessment

OVERALL VALIDATION RESULT:

Status: ☒ PASS

COMPONENT RESULTS:

HL (Boundary Layer Depth):	☒ EXCELLENT
Max difference:	0.012891 m
Mean relative error:	0.0123%
VISC_AZ:	☒ EXCELLENT
Median relative error:	0.000000%
DIFF_KZ_S:	☒ EXCELLENT
Median relative error:	0.037038%
DIFF_KZ_T:	☒ EXCELLENT
Median relative error:	0.037038%
GHAT:	☒ EXCELLENT
Median relative error:	0.012150%

PROVENANCE:

UUID Match:	☒ Yes
MITgcm UUID:	6d56ff34-f8ff-4406-81ee-d73f5f3d...
Python UUID:	6d56ff34-f8ff-4406-81ee-d73f5f3d...

ACCEPTANCE CRITERIA:

☒ EXCELLENT:	HL < 0.1m, mixing < 0.1% error
☒ GOOD:	HL < 1.0m, mixing < 1.0% error
☐ CHECK:	Exceeds thresholds - investigate